

Umang Saluja

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Profile

Computer Science undergraduate with experience building Machine Learning, NLP, Computer Vision, and Retrieval-Augmented Generation applications. Skilled in Python, TensorFlow, Scikit-learn, LangChain, Hugging Face, FastAPI, Flask, and vector databases, with AWS certifications spanning Machine Learning, AI, Cloud, and Solution Architecture.

Education

KIET Group of Institutions, Ghaziabad 2024–2028 B.Tech in Computer Science CGPA: 8.61/10.0 Till Semester III

Technical Skills

Languages: Python, Java, C

Machine Learning: Scikit-learn, TensorFlow, Keras, Feature Engineering, Model Evaluation, Classification, Regression, Hyperparameter Tuning, Model Optimization

Deep Learning: ANN, CNN, Transfer Learning, Neural Network Optimization

NLP and LLMs: LangChain, RAG, Hugging Face Transformers, NLTK, Prompt Engineering

Computer Vision: OpenCV, Image Processing, Food Image Classification

Backend and Databases: FastAPI, Flask, REST APIs, ChromaDB, Firebase Firestore, MongoDB

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Certifications

AWS Certified Machine Learning Engineer – Associate | AWS Certified AI Practitioner | AWS Certified Solutions Architect – Associate | AWS Certified Cloud Practitioner NPTEL Introduction to Machine Learning | NPTEL Data Analytics with Python | Flask

Relevant Coursework

Machine Learning, Artificial Intelligence, Probability and Statistics, Data Analytics, Advanced Data Structures, Database Systems, Operating Systems, Object-Oriented Programming

Selected Projects

Flutter Documentation RAG Pipeline

GitHub

LangChain, ChromaDB, TinyLlama, Hugging Face, Python, Google Colab

- Built a Retrieval-Augmented Generation pipeline for Flutter documentation using LangChain, TinyLlama-1.1B-Chat, and ChromaDB.
- Implemented web scraping, chunking, embedding generation, semantic retrieval, and local LLM inference workflows.
- Generated embeddings using all-MiniLM-L6-v2 and stored vectors in ChromaDB for similarity-based retrieval.
- Evaluated retrieval quality using different embedding and chunking strategies to improve contextual relevance.
- Conducted experiments on retrieval and generation workflows to improve answer quality and retrieval performance.

NutriLens – Smart Food Nutrition Analyzer

GitHub

Flutter, Flask, Firebase, CNN, Food-101, OpenCV

- Developed an AI-powered nutrition tracking application using Flutter, Flask, Firebase, and Computer Vision techniques.
- Trained a Convolutional Neural Network on the Food-101 dataset for food image classification.
- Designed REST APIs for image upload, prediction handling, and nutrition data retrieval.
- Integrated Firebase Authentication and Firestore for user management and meal history tracking.

AI Interview Coach App

GitHub

Flutter, FastAPI, Python, Scikit-learn, NLTK, Firebase

- Built an AI-powered interview preparation platform using Flutter, FastAPI, Scikit-learn, and NLP techniques.
- Implemented TF-IDF keyword relevance analysis, sentiment analysis, filler-word detection, and confidence-based feedback generation.
- Developed FastAPI services for response evaluation and structured feedback delivery.
- Integrated Firebase for authentication, response storage, and interview session management.

Achievements

- Solved 320+ Data Structures and Algorithms problems on LeetCode using Java, including 103 Medium and 12 Hard problems.
- Achieved an 88-day coding streak while consistently practicing problem-solving and algorithmic thinking.
- AWS Certified Machine Learning Engineer – Associate.
- Member, DSDL Machine Learning Domain; Member, Innogeeks Android Domain; Participant, Innogeeks Winter of Code 3.0.
- Secured First Prize in National Mathematics Day Project Presentation.